

Predictive Multiplicity in Classification

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ABSTRACT

In the context of machine learning, a prediction problem exhibits *multiplicity* if there exist several “good” models that attain identical or near-identical performance (i.e., accuracy, AUC, etc.). In this paper, we study the effects of multiplicity in human-facing applications, such as credit scoring and recidivism prediction. We introduce a specific notion of multiplicity – *predictive multiplicity* – to describe the existence of good models that output conflicting predictions. Unlike existing notions of multiplicity (e.g., the Rashomon effect), predictive multiplicity reflects irreconcilable differences in the predictions of models with comparable performance, and presents new challenges for common practices such as model selection and local explanation. We propose measures to evaluate the predictive multiplicity in classification problems. We present integer programming methods to compute these measures for a given datasets by solving empirical risk minimization problems with discrete constraints. We demonstrate how these tools can inform stakeholders on a large collection of recidivism prediction problems. Our results show that real-world prediction problems often admit many good models that output wildly conflicting predictions, and support the need to report predictive multiplicity in model development.

KEYWORDS

multiplicity, classification, integer programming, recidivism prediction, measurement

1 INTRODUCTION

Many methods in machine learning are designed to fit the *best* predictive model. In modern empirical risk minimization, for example, we aim to choose a model that optimizes an objective function (e.g., expected error) from a set of models that satisfy a given set of constraints (e.g., linear classifiers with equal FPR between groups). Assuming that stakeholders agree on this formulation, and we have the tools to solve it, we would still face a key issue in this approach: *there may be more than one “best” model*.

In machine learning, this notion – that a prediction problem may not have one “best” model but many “good” models that attain near-identical performance – is called *multiplicity*. Several works mention that prediction problems can exhibit multiplicity [see 30, 32], but only few discuss its practical implications, or provide concrete recommendations. The work of Breiman [7] is a major exception. In a seminal position paper, Breiman describes how multiplicity challenges the inherent validity of models: *if there exists several equally good models for a given dataset, each of which provides a different explanation of the data-generating process, then how can we tell which one is correct?* Drawing parallels between the

discordant explanations of good models and the discordant testimonies of witnesses in the motion picture “Rashomon,” Breiman calls this dilemma the *Rashomon effect*. In the context of his paper, the Rashomon effect is – in fact – an argument against the misuse of predictive models in scientific applications. Seeing how prediction problems can exhibit multiplicity, a cautious scientist should not draw scientific conclusions by “explaining” a predictive model, at least until there is evidence against the existence of multiplicity.

Machine learning has changed fundamentally since Breiman first discussed the Rashomon effect. Many models are now built for the sole purpose of making accurate predictions [20, 41]. In *human-facing applications*, such as credit scoring and recidivism prediction, these predictions affect people [6]. In turn, multiplicity challenges key aspects of how we build, interact, and trust models. Consider the following examples:

Recidivism Prediction: Say we are given a recidivism prediction model that performs well, but there exists a “competing model” that performs just as well. If these models output different predictions, then some individuals who are predicted to recidivate by the model we deploy could be predicted not to recidivate by a model that is equally valid. In this case, competing models with conflicting predictions present irreconcilable differences in the predictions of individuals [similar to the impossibility results in e.g., 9, 10, 21, 25]. Ignoring the existence of competing models would mean that we would arbitrate conflicting predictions by brute luck. In the worst case, a malicious practitioner could exploit this degree of freedom to choose a model that optimizes a secondary objective (e.g., a competing model that is least accurate on a certain group).

Credit Scoring: Say we must explain the prediction of a credit scoring model to a person [by producing e.g., a counterfactual explanation 27, 46]. If there exists several models that performed equally well, models with conflicting predictions would yield contradictory explanations for certain individuals. In turn, it would be misguided to trust the explanations of a model for a person who is assigned a conflicting prediction by a competing model. In this case, awareness of competing models with conflicting predictions would mitigate unwarranted rationalization of the model (e.g., via *fairwashing* [1] or *explanation bias* [19, 23]).

Stakeholders in modern applications – i.e., practitioners, policy-makers, decision-subjects – are broadly unaware of the fact that there may exist alternative models that output wildly different predictions from those that are deployed. One possible reason for this is the hypothesis that “good” models are “good” because they output similar predictions [see e.g., 29, for a critique]. We believe that it is important not only to raise awareness that competing models can produce conflicting predictions, but also to develop the tools to test this claim [see e.g. 22]. The latter goal is crucial since the incidence and severity of conflicting predictions inherently depends

on elements of model development such as the training data and model class. In such cases, presenting stakeholders with salient information on the existence and severity of conflicting predictions allows them to challenge key aspects of modern machine learning [as per the principles of contestability 17, 22] – from model selection (e.g., how should we choose between competing models, if we should choose one at all, and who should have a say in this decision) to downstream processes that assume we have chosen the best model (e.g., model validation, model regulation, explanation).

In this paper, we introduce a general framework to evaluate the multiplicity of predictions in classification problems. Our objective is to provide stakeholders with information on the incidence and severity of multiplicity. We introduce three measures to capture how predictions vary over the set of all good models:

Individual Ambiguity: Does there exist a good model that produces a conflicting prediction for a given person?

Overall Ambiguity: How many individuals can be assigned a different prediction by at least one good model?

Maximum Discrepancy: What is the maximum number of conflicting predictions between a baseline model and any good model?

These are model-agnostic measures that capture meaningful aspects of how multiplicity affects individuals in general classification problems (binary or multiclass).

To compute these measures, one must find specific models within a set of “good” models (e.g., a model that maximizes discrepancy while achieving comparable accuracy with respect to a baseline model we would deploy). We present tools to compute these quantities exactly for linear classification problems, then apply our tools to recidivism prediction problems where predictive multiplicity has important ethical implications.

Contributions. The main contributions of our paper include:

1. We present formal measures of how multiplicity affects individual prediction in classification tasks.
2. We develop new integer programming tools to efficiently compute these measures for linear classification problems.
3. We apply our tools to study the incidence and severity of multiplicity in recidivism prediction. Our results show that real-world datasets can often admit several good models that exhibit a large number of conflicting predictions.¹
4. We discuss how multiplicity can affect key aspects of modern machine learning applications. Specifically, we discuss how the models produced using our tools can support common practices such as addressing performance disparities between protected groups and calibrating trust in explanations.
5. We provide a software implementation of our tools in Python at <https://github.com/ustunb/pm>

¹In the ProPublica COMPAS dataset [3], for example, we find that 44% of predictions can be changed by some model within 1% training and test error of the baseline model. Further, we find a single model that changes 17% of predictions from the optimal model and only exhibits a 1% increase in training and test error. (see Section 4)

(x_1, x_2)	LABELS		CLASSIFIERS			
	n^+	n^-	$\hat{h}_a$	$\hat{h}_b$	$\hat{h}_c$	$\hat{h}_d$
(0, 0)	0	100	+	−	−	−
(0, 1)	100	0	+	−	+	+
(1, 0)	100	0	+	+	−	+
(1, 1)	0	100	−	−	−	+

Figure 1: Toy example showing how multiple classifiers can attain identical performance while producing conflicting predictions. We train linear classifiers on a dataset with 400 examples. Here, $\hat{h}_a, \hat{h}_b, \hat{h}_c, \hat{h}_d$ denote the linear classifiers that attain the minimal training error (25%), and n^+ and n^- denote the number of examples with $y = +1$ and $y = -1$ for a given value of $(x_1, x_2) \in \{0, 1\}^2$. The predictions of any 2 classifiers differ on 200 points (i.e., 50% of the training sample). Thus, the maximum discrepancy is 50%. Moreover, there exists a classifier that can predict +1 or −1 for each point. Thus, the overall ambiguity is 100%.

1.1 Related Work

Multiplicity. The potential for predictive models to exhibit multiplicity has been known since at least the 1980s [see e.g., 32]. McCullagh and Nelder [30], for example, write “data will often point with almost equal emphasis on several possible models, and it is important... [to] recognize and accept this.”

Breiman [7] presents an early discussion of the practical implications of multiplicity. He argues that the existence of alternative models undermines the validity of using a single model to support broader claims about the data-generating mechanism (e.g., conflating the important variables of a recidivism prediction instrument with the root causes of criminal behavior). Breiman does not advocate a prescription to resolve multiplicity. Instead, he cautions against “automatic methods of model selection” and warns that “the multiplicity problem and its effect on conclusions drawn from models needs serious attention.”

Recent work in machine learning tackles multiplicity from the “Rashomon” perspective. Fisher et al. [13] and Dong and Rudin [12], for example, present techniques to evaluate variable importance over the set of good models. Semenova and Rudin [40] present a new measure of the relative size of the set of good models, and show that it can characterize prediction problems where simple models tend to perform well.

Our work differs from prior work on multiplicity in that we explicitly focus on settings where good models produce conflicting predictions. This kind of multiplicity implies irreconcilable differences between subsets of predictions, [similar to impossibility results in e.g., 9, 10, 21, 25].

Model Selection. Technical solutions for how to “resolve” multiplicity broadly consists of techniques for tie-breaking or reconciliation. Traditional approaches to model selection often seek to choose a single model that is “closest” to a “true model” by breaking ties on the basis of estimators such as the AIC, BIC, or K-CV error [see e.g., 29, for a critique]. Bayesian approaches explicitly represent uncertainty over the set of competing models by means of a posterior

probability distribution, but then use the posterior distribution to construct a single model for deployment [e.g., via a majority vote or randomization procedure 15, 28, as in PAC Bayesian approaches].

While both approaches aim to improve out-of-sample performance, they may not be appropriate to handle all incidences of multiplicity. In effect, there may exist cases where tie-breaking between models may never achieve its intended goal of improving out-of-sample performance (e.g. in Figure 1). More broadly, choosing a single model in an automated fashion would overlook important aspects of multiplicity in human-facing applications, such as how to resolve individuals with ambiguous predictions.

An alternative approach aims to choose a single model to “break ties” between equally good models on the basis of secondary criteria, such as simplicity [40] or operational cost [42].

2 FRAMEWORK

In this section, we introduce measures to describe the multiplicity of predictions in classification, and present formal guarantees relating multiplicity to training accuracy. We include proofs for all results in Appendix A.

For the sake of clarity, we will define our measures for classification problems where we care about accuracy. We note, however, that our measures apply to general settings where we care about other performance metrics (e.g., AUC, TPR), specific types of constraints [e.g., fairness constraints 8, 11, 48], or must predict more than two outcomes.

2.1 Preliminaries

We consider a standard binary classification problem. We are given a dataset of n examples $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where each *example* consists of a vector of $d + 1$ features $\mathbf{x}_i = (1, x_{i1}, x_{i2}, \dots, x_{id}) \in \mathbb{R}^{d+1}$ and a label $y_i \in \{-1, +1\}$.

Our goal is to train the most accurate classifier $h : \mathbb{R}^{d+1} \rightarrow \{-1, +1\}$ from a fixed hypothesis class $\mathcal{H}$ that optimizes the true risk (i.e., test error). To this end, we minimize the training error (i.e., 0-1 loss):

$$\hat{R}(h) := \sum_{i=1}^n \mathbb{1}[h(\mathbf{x}_i) \neq y_i].$$

We denote the optimal classifier as

$$\hat{h} \in \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}(h).$$

We note that there may exist several optimal classifiers.

Level Sets. We measure the incidence and severity of multiplicity among the set of *good models* – i.e., the subset of classifiers in $\mathcal{H}$ that attain optimal or near-optimal performance. We refer to the set of all models whose performance differs by at most ϵ as the ϵ -level set. Since we use the error rate to measure performance, we sometimes refer to ϵ as the *error tolerance*. We note, however, that level sets can be defined using other metrics [see e.g., 12, 13, for general level sets of models].

DEFINITION 1 (ϵ -LEVEL SET). *Given any classifier h and a fixed hypothesis class $\mathcal{H}$, the ϵ -level set of h is the collection of all models $h' \in \mathcal{H}$ that make at most $\hat{R}(h) + \epsilon$ mistakes over the training data:*

$$S_\epsilon(h) := \{h' \in \mathcal{H} : |\hat{R}(h') - \hat{R}(h)| \leq \epsilon\}.$$

In general, it is possible for a prediction problem to admit several models that attain identical performance but produce conflicting predictions (see e.g., Figure 1). In spite of this, there is good reason to measure multiplicity over level sets where $\epsilon > 0$. It is well-known that a model that minimizes a performance on the training set may not minimize performance in deployment. In fact, competing models – i.e., models that attain near-optimal performance on the training set – typically perform better than the optimal model in deployment (see e.g., Section 4 for empirical evidence). In such settings, it would not be defensible to rule out near-optimal models on the basis of small differences in training error.

In practice, one should therefore consider a set of models that are plausibly contain a model that will perform the best in deployment. That is, we should set ϵ to be large enough so the ϵ -level set is likely to contain the model that optimizes out-of-sample performance.² In our experiments, we measure the differences in predictions for all possible values of ϵ . We find that small values of ϵ can produce strikingly large differences in predictions, and that this effect can occur even when $\epsilon = 0$.

2.2 Predictive Multiplicity

We say that a prediction problem exhibits *predictive multiplicity* if there exist several models that attain similar performance but produce conflicting predictions on the training data.

DEFINITION 2 (PREDICTIVE MULTIPLICITY). *Given a training dataset D and a fixed hypothesis class $\mathcal{H}$, we say that a problem exhibits predictive multiplicity over the ϵ -level set $S_\epsilon(h)$ if there exists two models $h, h' \in S_\epsilon(h)$ such that $h(\mathbf{x}_i) \neq h'(\mathbf{x}_i)$ for some $\mathbf{x}_i \in D$.*

Predictive multiplicity differs from general multiplicity in that it ignores cases where a problem exhibits multiplicity due to the existence of several models that produce the same predictions (see e.g., Figure 2 for an example). Thus, any incidence of predictive multiplicity implies that there exists two or more models that produce irreconcilable predictions on the training data.

2.3 Measures of Predictive Multiplicity

Maximum Discrepancy. Given a baseline classifier $\hat{h}$, we measure the severity of predictive multiplicity by counting the number of conflicting predictions between $\hat{h}$ and a model in the ϵ -level set. We refer to the number of predictions that differ between $\hat{h}$ and specific model h as the *discrepancy*, denoted $\delta(\hat{h}, h)$. Formally, the discrepancy is

$$\delta(\hat{h}, h) := \#\{\mathbf{x}_i \in X : \hat{h}(\mathbf{x}_i) \neq h(\mathbf{x}_i)\}.$$

We denote the maximum discrepancy over the set of models in the ϵ -level set as δ_ϵ .

DEFINITION 3 (MAXIMUM DISCREPANCY). *Given a training dataset D , a classifier h , and the ϵ -level set around h , the maximum discrepancy of h over the ϵ -level set is:*

$$\delta_\epsilon(h) := \max_{h' \in S_\epsilon(h)} \#\{\mathbf{x}_i \in X : h'(\mathbf{x}_i) \neq h(\mathbf{x}_i)\}.$$

²This value can be determined in a principled way by applying generalization bounds, by constructing confidence intervals of out-of-sample performance (e.g., via bootstrapping, or cross-validation).

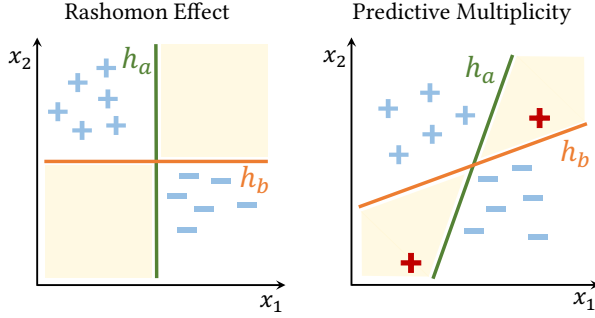


Figure 2: Models that attain the same performance may not necessarily output the same predictions. On the left, we depict a pair of classifiers h_a and h_b that output the same predictions, but that produce conflicting explanations of the relative importance of x_1 vs. x_2 (i.e., the Rashomon effect). On the right, we depict a pair of classifiers h_a and h_b that output conflicting predictions (i.e., predictive multiplicity). Regions of conflict are shown in yellow, and points of conflict are highlighted in red. All classifiers optimize accuracy on their respective datasets. Prediction problems can exhibit both types of multiplicity, but predictive multiplicity requires that there is hard evidence of irreconcilable differences between their predictions.

If the maximum discrepancy is small, it means that near-optimal models must output similar predictions, reflecting the common intuition that similarly performing models make similar prediction. If the maximum discrepancy is large, this means there exists a near-optimal solution which makes very different predictions than the baseline classifier. Such a result signals inherent trade-offs between predictions, possibly resulting from noisy data or structure in the data the model class cannot fit. Large discrepancies also means practitioners should consider choosing between models on the basis of secondary objectives, and rule out the possibility of adversarial model selection.

Individual Ambiguity. If there exists a classifier in the ϵ -level set making a different prediction on $\mathbf{x}_i$, then the prediction $\hat{h}(\mathbf{x}_i)$ is susceptible to model choice. In this case, we say that the prediction on $\mathbf{x}_i$ is *ambiguous*. This has important practical implications, which we illustrate in Section 4. For one, it may provide grounds for an individual given a negative prediction to contest their prediction. Additionally, generating local explanations of the data using $\hat{h}(\mathbf{x}_i)$ may be suspect when there exists an alternative model h with similar error such that $h(\mathbf{x}_i) \neq \hat{h}(\mathbf{x}_i)$.

Overall Ambiguity. We introduce the *ambiguity* of a model as the number of instances which exhibit individual ambiguity. While maximum discrepancy shows a practitioner the “most different” model, overall ambiguity measures how many individuals are subject to manipulation of their predictions.

DEFINITION 4 (AMBIGUITY). We say that a classifier $h \in \mathcal{H}$ exhibits an ambiguity of k if there are k individuals $\mathbf{x}_i \in X$ for which there exists an alternative model $h' \in S_\epsilon(h)$ such that the predictions

of h and h' on $\mathbf{x}_i$ differ. Formally, we define the ambiguity of h to be

$$\alpha_\epsilon(h) = \#\{\mathbf{x}_i \in X : \exists h' \in S_\epsilon(h) \text{ s.t. } h'(\mathbf{x}_i) \neq h(\mathbf{x}_i)\}$$

Ambiguity and maximum discrepancy capture salient aspects of predictive multiplicity in a given prediction problem. We note that these quantities change based on the choice of baseline model. In turn, we view the ambiguity and maximum discrepancy of the best performing model as canonical measures of predictive multiplicity for a given problem, and denote them as α_ϵ and δ_ϵ , respectively. We observe that $\delta_\epsilon \leq \alpha_\epsilon$, since the predictions flipped by a single model are a subset of those flipped by any model. Since different predictions may be flipped by different models in the level set, there may be cases where ambiguity far exceeds the maximum discrepancy. This indicates an unintuitive scenario where all good models are similar, but many predictions are affected by model choice. In other cases, we find that ambiguity and maximum discrepancy track closely to one another. This indicates that a single model is capable of expressing most of the predictive multiplicity.

2.4 Formal Guarantees

We conclude this section with simple bounds that illustrate the relationship between training accuracy and predictive multiplicity.

In Proposition 2.1, we bound the number of distinct prediction profiles produced by models in the ϵ -level. In Proposition 2.2, we bound the number of conflicts between the optimal model and any single model in the ϵ -level set.

PROPOSITION 2.1 (BOUND ON THE NUMBER OF GOOD MODELS). Let $\underline{\epsilon} = \max(-\epsilon, -\hat{R}(\hat{h}))$ and $\bar{\epsilon} = \min(\epsilon, n - \hat{R}(\hat{h}))$ denote the effective lower and upper bounds of the ϵ -level set respectively, since the number of mistakes of any model must be in the interval $[0, n]$. The number of classifiers in the ϵ -level set of the optimal classifier is bounded by

$$|S_\epsilon(\hat{h})| \leq \sum_{i=\underline{\epsilon}}^{\bar{\epsilon}} \binom{n}{\hat{R}(\hat{h}) + i}$$

PROPOSITION 2.2 (BOUND ON THE MAXIMUM DISCREPANCY). The maximum number of conflicts between $\hat{h}$ and an alternative model $h \in S_\epsilon(\hat{h})$ is bounded by

$$\delta_\epsilon \leq 2\hat{R}(\hat{h}) + \epsilon$$

Propositions 2.1 and 2.2 demonstrate how the severity of predictive multiplicity changes based on the accuracy of a baseline model. These bounds illustrate the “pinching” effect of accuracy on multiplicity. As the number of mistakes decreases, the maximum number of possible trade-offs between conflicting points also decreases. In other words, there is more “room” for predictive multiplicity in settings where models perform poorly.

3 METHODOLOGY

In this section, we present integer programming methods to evaluate multiplicity in linear classification problems.

Overview. Our methods aim to compute the values of discrepancy and ambiguity in Section 2 by training three kinds of linear classifiers:

1. $\hat{h}_1$: a pathological classifier that minimizes training error subject to the constraint that it must flip the prediction of $\hat{h}$ on the

example $\mathbf{x}_i$. We will use this classifier to determine the individual ambiguity of i . By training $\hat{h}_i$ for all points $i \in \{1, \dots, n\}$, we will then be able to compute overall ambiguity.

2. $\hat{h}_\delta$: a classifier in the ϵ -level set that maximizes the number of conflicting predictions. This classifier will be used to compute maximum discrepancy.
3. $\hat{h}$: a baseline classifier that minimizes the training error. While we can compute ambiguity and discrepancy using any baseline classifier, computing them with respect to $\hat{h}$ provides an uncontested account of multiplicity since it reflects the best possible model that one could build.³

We train each classifier by solving an empirical risk minimization problem with discrete constraints. We formulate each problem as an mixed integer program (MIP), which can be solved using an IP solver such as CPLEX, Gurobi, or CBC. Solving an IP produces: (i) the objective value (i.e., the training error of the best model); (ii) a lower bound on the objective value; (iii) the coefficients of a linear classifier.

3.1 Finding the Most Accurate Classifier

We can train a classifier that directly minimizes the 0-1 loss on the training data by solving the following optimization problem:

$$\min_{h \in \mathcal{H}} \sum_{i=1}^n \mathbb{1}[h(\mathbf{x}_i) \neq y_i] \quad (1)$$

We train a linear classifier $h(\mathbf{x}_i) = \mathbf{w}^\top \mathbf{x}_i$ that optimizes the 0-1 loss by solving the MIP formulation:

$$\min \sum_{i=0}^n l_i \quad (2a)$$

$$\text{s.t.} \quad M_i l_i \geq y_i(\gamma - \sum_{j=0}^d w_j x_{ij}) \quad i = 1, \dots, n \quad (2a)$$

$$1 = l_i + l_{i'} \quad (i, i') \in K \quad (2b)$$

$$w_j = w_j^+ + w_j^- \quad j = 0, \dots, d \quad (2c)$$

$$1 = \sum_{j=0}^d (w_j^+ - w_j^-) \quad (2d)$$

$$l_i \in \{0, 1\} \quad i = 1, \dots, n$$

$$w_j \in [-1, 1] \quad j = 0, \dots, d$$

$$w_j^+ \in [0, 1] \quad j = 0, \dots, d$$

$$w_j^- \in [-1, 0] \quad j = 0, \dots, d$$

Here, constraints (2a) set the mistake indicators $l_i \leftarrow \mathbb{1}[h(\mathbf{x}_i) \neq y_i]$. These constraints depend on a margin parameter $\gamma > 0$, which should be set to a small positive number (e.g., 10^{-4}), and “Big-M” parameters M_i which can be set as $M_i = \gamma + \max_{\mathbf{x}_i \in X} \|\mathbf{x}_i\|_\infty$ since we have fixed $\|\mathbf{w}\|_1 = 1$ in constraint (2d). Constraint (2b) produces an improved lower bound by encoding the necessary condition that any classifier must make exactly one mistake between any two points $(i, i') \in K$ with identical features $\mathbf{x}_i = \mathbf{x}_{i'}$ but conflicting

labels. Here, $K = \{(i, i') : \mathbf{x}_i = \mathbf{x}_{i'}, y_i = +1, y_{i'} = -1\}$ is the set of points with conflicting labels.

3.2 Computing Maximum Discrepancy

Given a training dataset and a baseline classifier $\hat{h}$, we can compute the maximum discrepancy (Definition 3) for a given value of ϵ by training the classifier in the ϵ -level set that maximizes discrepancy with respect to the baseline classifier.

We denote this classifier as $\hat{h}_\delta$, and recover it by solving the following optimization problem:

$$\begin{aligned} \min_{h \in \mathcal{H}} \quad & \sum_{i=1}^n \mathbb{1}[h(\mathbf{x}_i) \neq \hat{h}(\mathbf{x}_i)] \\ \text{s.t.} \quad & |\hat{R}(h) - \hat{R}(\hat{h})| \leq \epsilon \end{aligned} \quad (3)$$

This optimization problem in (3) minimizes the agreement between $\hat{h}_\delta$ and $\hat{h}$, which is equivalent to maximizing discrepancy. This problem can be viewed as an empirical risk minimization problem where we train a model to predict the “labels” $\{-\hat{h}(\mathbf{x}_i)\}_{i=1}^n$.

We train a linear classifier $\hat{h}_\delta(\mathbf{x}_i) = \mathbf{w}^\top \mathbf{x}_i$ that maximizes discrepancy with the baseline classifier by solving the MIP formulation:

$$\max \sum_{i=0}^n \delta_i \quad (4a)$$

$$\text{s.t.} \quad M_i(1 - \delta_i) \geq \gamma + \hat{h}(\mathbf{x}_i) \sum_{j=0}^d w_j x_{ij} \quad i = 1, \dots, n \quad (4a)$$

$$M_i l_i \geq y_i(\gamma - \sum_{j=0}^d w_j x_{ij}) \quad i = 1, \dots, n \quad (4b)$$

$$\epsilon \geq \sum_{i=1}^n \delta_i(1 - 2l_i) \quad (4c)$$

$$1 = l_i + l_{i'} \quad (i, i') \in K \quad (4d)$$

$$w_j = w_j^+ + w_j^- \quad j = 0, \dots, d \quad (4e)$$

$$1 = \sum_{j=0}^d (w_j^+ - w_j^-) \quad (4f)$$

$$\delta_i \in \{0, 1\} \quad i = 1, \dots, n$$

$$l_i \in \{0, 1\} \quad i = 1, \dots, n$$

$$w_j \in [-1, 1] \quad j = 0, \dots, d$$

$$w_j^+ \in [0, 1] \quad j = 0, \dots, d$$

$$w_j^- \in [-1, 0] \quad j = 0, \dots, d$$

Here, the objective is to maximize $\sum_{i=0}^n \delta_i$, where δ_i indicates whether the prediction on $\mathbf{x}_i$ differs from the optimal classifier. Constraints (4a) set the discrepancy indicators $\delta_i \leftarrow \mathbb{1}[h(\mathbf{x}_i) \neq \hat{h}(\mathbf{x}_i)]$. Constraint (4c) enforces that h is in the ϵ -level set, i.e. that $|\hat{R}(h) - \hat{R}(\hat{h})| \leq \epsilon$. All other constraints are adopted from the baseline MIP formulation. For the maximum discrepancy MIP, the objective value is the maximum discrepancy found by the solver in the ϵ -level set. The upper bound is the maximum possible discrepancy. When the objective value equals the upper bound, the solution provably maximizes the number of conflicts with the baseline classifier.

Given a baseline classifier $\hat{h}$ that is provably optimal, we can slightly reduce training time by adding the constraint that $\sum_{i=1}^n \delta_i(1 -$

³When we compute our measures with respect to the best baseline, we rule out the possibility that multiplicity arises due to a “suboptimal” model. In such cases, the only way to address multiplicity is to change the original problem formulation (e.g., by collecting more data, or incorporating additional objective or constraints to address secondary objectives [see 36]).

$2l_i) \geq 0$. This constraint reflects the fact that we cannot increase the error rate by using an optimal baseline.

Computing Maximum Discrepancy for all ϵ . While there are systematic ways to set the error tolerance, we can avoid the need to choose ϵ *a priori*, by computing discrepancy for all possible values of ϵ .

In Algorithm 1, we present a simple procedure to determine the value of maximum discrepancy for all possible values of ϵ . The procedure starts by solving the MIP in 4 for $\epsilon=0$, and then proceeds to solve the MIPs in 4 for increasing values of ϵ . At each iteration, it uses the solutions produced from previous iterations to initialize the current MIP. This process is many times faster than solving the MIP separately for various error tolerances.

Algorithm 1 Compute the maximum discrepancy path over ϵ

Input

IP instance of IP (4)
 $\hat{h}$ baseline classifier
 $\underline{\epsilon}, \bar{\epsilon}$ min and max ϵ to consider

Initialize

$\mathcal{P} \leftarrow \emptyset$ maximum discrepancy path
 $\mathcal{S} \leftarrow \emptyset$ solution pool

for ϵ in $\{\underline{\epsilon}, \underline{\epsilon} + 1, \dots, \bar{\epsilon}\}$ **do**

 IP $\leftarrow$ initialize IP with optimal solution from $\mathcal{S}$

$\hat{h}_\delta \leftarrow$ solve IP with error tolerance ϵ

$\delta_\epsilon \leftarrow$ number of conflicts between $\hat{h}_\delta$ and $\hat{h}$

$\mathcal{P} \leftarrow \mathcal{P} \cup (\epsilon, \delta_\epsilon)$

$\mathcal{S} \leftarrow \mathcal{S} \cup (\hat{h}_\delta)$

end for

Output: $\mathcal{P}$ the maximum discrepancy for each ϵ considered

3.3 Computing Ambiguity

Given a training dataset and a baseline classifier $\hat{h}$, we compute the value of overall ambiguity in (Definition 4) for all possible values of ϵ by training a pathological classifier $\hat{h}_i$ for each point $i = \{1, \dots, n\}$.

Given the baseline classifier $\hat{h}$ and the instance $\mathbf{x}_i \in X$, the pathological classifier for point i $\hat{h}_i$ is the most accurate linear classifier that outputs a conflicting prediction on point i . We can train $\hat{h}_i$ by solving an optimization problem of the form:

$$\begin{aligned} \min_{h \in \mathcal{H}} \quad & \sum_{k=1}^n \mathbb{1}[h(\mathbf{x}_k) \neq y_i] \\ \text{s.t.} \quad & h(\mathbf{x}_i) \neq \hat{h}(\mathbf{x}_i) \end{aligned} \quad (5)$$

Here, $h(\mathbf{x}_i) \neq \hat{h}(\mathbf{x}_i)$ is a constraint that forces any feasible classifier h to assign $\mathbf{x}_i$ a prediction that is in conflict with $\hat{h}(\mathbf{x}_i)$. Seeing how this optimization problem is identical to the optimization problem in (1), we train this model using a variant of MIP formulation in (2) that contains all of the same variables and constraints, and that includes the constraint of the form:

$$\gamma \leq -\hat{h}(\mathbf{x}_i) \sum_{j=0}^d w_j x_{ij} \quad (6)$$

The solution to this MIP returns the coefficients and training error of the classifier $\hat{h}_i$. With these quantities, we can determine if point i exhibits individual ambiguity over the set of ϵ -good models by verifying if the following inequality holds:

$$\hat{R}(\hat{h}_i) \leq \hat{R}(\hat{h}) + \epsilon.$$

Given a set of pathological classifiers $\hat{h}_i$ for each point $i = \{1, \dots, n\}$, we can determine the overall ambiguity over the set of ϵ -good models as:

$$\alpha_\epsilon = \#\{\mathbf{x}_i \in X : \hat{R}(\hat{h}_i) \leq \hat{R}(\hat{h}) + \epsilon\} \quad (7)$$

The expression in (7) states that the overall ambiguity of a prediction task over the ϵ -level set is equivalent to the number of pathological classifiers with -optimal training accuracy.

Speed-Ups. Our proposed approach to evaluate overall ambiguity has the benefit that it can evaluate this value for all possible values of ϵ . In practice, however, this also requires that we train the best possible pathological classifier for every instance $\mathbf{x}_i$, which may involve substantial computation.

In our implementation, we aim to reduce the computation involved with this procedure in several ways:

- We initialize the lower bound of the MIP that we use to train $\hat{h}_i$ to the lower bound of the baseline MIP (since the baseline the solution space is a subset of the solution space of the baseline MIP).
- Since exactly one of $\hat{h}$ and $\hat{h}_i$ must make a mistake on $\mathbf{x}_i$, we can also add the constraint that $l_i = \mathbb{1}[\hat{h}(\mathbf{x}_i) = y_i]$.
- When training many $\hat{h}_i$ classifiers, we find a large speed-up from maintaining a solution pool of all trained MIPs and initializing to the best existing solution h which satisfies $h(\mathbf{x}_i) \neq \hat{h}(\mathbf{x}_i)$.

3.4 Discussion

Our use of integer programming is motivated by several reasons:

- Integer programming allows us to directly optimize and constrain the exact quantities that we care about. In effect, integer programming trains models in a way that is guaranteed to consider *all* possible models within an ϵ -level set. In contrast, standard approximations (e.g., the use of a convex loss function) may fail to uncover models, and thus underestimate the true ambiguity and discrepancy.
- Although integer programming is computationally intractable, modern integer programming tools allow us to solve empirical risk minimization for many real-world problems [see e.g., 5, 33, 43–45]. When integer programs cannot be solved, they will produce bounds that are useful. For example, in cases where the maximum discrepancy MIP does not solve to optimality, the MIP returns the best solution found and an upper bound on the maximum discrepancy that is possible.

Our formulations can be modified to optimize other objectives (e.g., class-based accuracy) and to handle various constraints (e.g., parity constraints). In addition, we can train a large class of “linearizable” rule-based models.⁴ Our measures can be evaluated without these

⁴These are rule-based models that can be expressed as special kinds of linear classifiers, such as decision sets [24], decision lists [2], and and-of-or rules [47].

tools, and provide useful information (albeit in a heuristic manner that may underestimate or overestimate these measures).

4 EXPERIMENTS

In this section, we present experiments where we apply our tools to evaluate predictive multiplicity in recidivism prediction problems.⁵ We have four goals: (i) to test the incidence and severity of multiplicity in real-world classification problems; (ii) to show how our tools can inform stakeholders in such settings; (iii) to discuss the implications of multiplicity with respect to transparency and accountability of such models; (iv) to show that we can evaluate multiplicity using existing tools. For the sake of reproducibility, we attach the software, scripts, and datasets necessary to reproduce all of the results to our submission.

4.1 Setup

Datasets. We define a set of 9 recidivism prediction problems from 3 studies of recidivism in the United States:

- Felony Defendants in Large Urban Counties [35] (*pretrial*);
- Recidivism of Prisoners Released in 1994 study [34] (*recidivism*);
- ProPublica COMPAS repository [3] (*compas*).

We use the raw data from each study to create training datasets by following common practices in the development of recidivism prediction instruments [see e.g., guidelines in 37]. Each dataset consists of a binary outcome (e.g., a proxy variable such as arrest, arrest for violent crime) and a set of features for prisoners released in a specific state (e.g., NY, CA). We process each dataset by dropping instances with missing entries, binarizing features, and oversampling the minority class to equalize the number of instances with positive and negative labels.⁶ We present an overview of the number of instances and features in each dataset in Table 1.

Measurement with our Tools. We evaluate the incidence and severity of predictive multiplicity for each of the datasets as follows. We first split each training dataset into a *training set* composed of 80% of points, and a *test set* composed of 20% of the points. For each dataset, we use the training data to fit a *baseline classifier* – i.e., the best classifier that we would deploy – by solving the IP in Section 3.1. Given the baseline classifier, we measure the maximum discrepancy and ambiguity among the set of good models by solving the IPs in Sections 3.3 and 3.2 for *all possible values* of the error tolerance ϵ . In order to measure maximum discrepancy, we solve MIP (4) for all possible values of the error tolerance $\epsilon \in [0, n/2]$. In order to measure overall ambiguity, we solve the MIP described in Section 3.3 for each distinct point in the training data.⁷

We solve all IPs on a Unix machine with 3.33 GHz CPU with 16 GB RAM. We allocate at most 6 hours to train the baseline model, 6 hours to train all maximum discrepancy models, and 6 hours to train the individual ambiguity models.

⁵We consider recidivism prediction *not* in an effort to build new recidivism prediction instruments, but because it is a domain where the existence of predictive multiplicity has serious ethical implications and may present an additional critique [see e.g., 4, 16, 26, for a critique].

⁶Oversampling ensures that we can directly minimize the training error without having to worry about producing trivial classifiers. This also allows us to report the error rate as opposed to class-specific error rates. We find that oversampling has a trivial effect on our measures of multiplicity ($< 1\%$).

⁷In practice, we note that these measures would be computed for a specific value of ϵ .

Dataset	n	d	Training Error	Test Error
compas_arrest	5,380	18	32.7%	33.4%
compas_violent	8,768	18	37.7%	37.9%
pretrial_CA_arrest	9,926	22	34.1%	34.4%
pretrial_CA_fta	8,738	22	36.3%	36.3%
pretrial_NY_fta	730	22	25.9%	29.9%
recidivism_CA_arrest	114,522	20	34.4%	34.2%
recidivism_CA_drug	96,664	20	36.3%	36.2%
recidivism_NY_arrest	31,624	20	31.0%	31.8%
recidivism_NY_drug	27,526	20	32.5%	33.6%

Table 1: Performance of baseline classifiers for all datasets. Here, n is the number of instances in the dataset and d is the number of features. We train each model by minimizing the 0-1 loss on the training data, which does not require parameter tuning. All 9 models generalize as shown training error is close to test error. This is expected given that we fit models from a simple hypothesis class on datasets $n \gg d$.

Measurement using Standard Methods. We also consider an alternative analysis where we compute our proposed measures over a large pool of linear classifiers. We include this analysis to show that a systematic evaluation of predictive multiplicity can still reveal salient information – even without using specialized techniques.

We construct a pool of linear classifiers via penalized logistic regression using the glmnet package of [14]. Each model is trained using logistic regression along with a weighted combination of the ℓ_1 and ℓ_2 penalties. Thus, the coefficients of each model are the solution to the optimization problem:

$$\min_{\mathbf{w} \in \mathbb{R}^{d+1}} \frac{1}{2n} \sum_{i=1}^n \log(1 + \exp(-\langle \mathbf{w}, y_i \mathbf{x}_i \rangle)) + \lambda \cdot \left(\alpha \|\mathbf{w}\|_1 + (1 - \alpha) \|\mathbf{w}\|_2^2 \right)$$

Here $\alpha \in [0, 1]$ is the elastic-net mixing parameter and $\lambda \geq 0$ is a regularization penalty. We train 1,100 models by choosing 1,100 combinations of (α, λ) : 11 values of $\alpha \in \{0.0, 0.1, \dots, 0.9, 1.0\} \times 100$ values of λ (chosen automatically by glmnet for each α). This produces 1,100 models that include models obtained by: (i) Lasso (ℓ_1 -penalty), which corresponds to when $\alpha = 1.0$; (ii) Ridge (ℓ_2 -penalty), which corresponds to when $\alpha = 0.0$; (iii) standard logistic regression, which corresponds to when $\alpha = 0.0$ and λ is small.

In this setting, we choose our baseline model as the model that minimizes the 5-fold cross-validation error. We then compute discrepancy and ambiguity by comparing the baseline model to models in the pool that attain at most ϵ additional training error.

4.2 Results

We find that all datasets exhibit predictive multiplicity. Our key experimental findings include:

- Predictive multiplicity appears with startling severity in important human-facing prediction tasks.
- Predictive multiplicity has important downstream effects. In some cases, the burden of multiplicity differs greatly between different subgroups.

	Baseline Model	Individual Ambiguity Model	Max Discrepancy Model
$h(x_p)$	+1	-1	-1
Error Train	32.7%	32.7%	33.6%
Error Test	33.4%	33.4%	34.5%
Discrepancy Train	0.0%	0.0037%	16.8%
Discrepancy Test	0.0%	0.0%	15.1%
Score Function	+ 0.5 $age \leq 25$	+ 10.3 $age \leq 25$	+ 7.7 $age \leq 25$
	+ 0.0 age_{25-45}	+ 0.0 age_{25-45}	+ 0.0 age_{25-45}
	- 16.4 $age \geq 46$	- 9.9 $age \geq 46$	- 7.8 $age \geq 46$
	- 16.3 $female$	- 9.7 $female$	- 7.6 $female$
	- 0.2 $n_priors=0$	+ 0.0 $n_priors=0$	- 7.8 $n_priors=0$
	- 0.1 $n_priors \geq 1$	+ 0.0 $n_priors \geq 1$	+ 0.0 $n_priors \geq 1$
	+ 16.4 $n_priors \geq 2$	+ 19.8 $n_priors \geq 2$	+ 7.4 $n_priors \geq 2$
	+ 16.6 $n_priors \geq 5$	+ 10.1 $n_priors \geq 5$	+ 7.8 $n_priors \geq 5$
	+ 0.0 $n_juvenile_misdemeanors=0$	+ 0.0 $n_juvenile_misdemeanors=0$	+ 0.0 $n_juvenile_misdemeanors=0$
	- 0.1 $n_juvenile_misdemeanors \geq 1$	- 0.1 $n_juvenile_misdemeanors \geq 1$	+ 0.1 $n_juvenile_misdemeanors \geq 1$
	+ 0.0 $n_juvenile_misdemeanors \geq 2$	- 10.1 $n_juvenile_misdemeanors \geq 2$	- 0.1 $n_juvenile_misdemeanors \geq 2$
	- 32.6 $n_juvenile_misdemeanors \geq 5$	- 9.5 $n_juvenile_misdemeanors \geq 5$	- 15.2 $n_juvenile_misdemeanors \geq 5$
	+ 0.0 $n_juvenile_felonies=0$	- 9.9 $n_juvenile_felonies=0$	+ 7.7 $n_juvenile_felonies=0$
	- 0.2 $n_juvenile_felonies \geq 1$	- 10.1 $n_juvenile_felonies \geq 1$	+ 0.0 $n_juvenile_felonies \geq 1$
	+ 0.3 $n_juvenile_felonies \geq 2$	+ 0.3 $n_juvenile_felonies \geq 2$	+ 15.4 $n_juvenile_felonies \geq 2$
	+ 0.0 $n_juvenile_felonies \geq 5$	+ 0.0 $n_juvenile_felonies \geq 5$	+ 0.0 $n_juvenile_felonies \geq 5$
	- 0.2 $charge_degree=M$	- 0.2 $charge_degree=M$	- 7.5 $charge_degree=M$
	+ 0.0	+ 0.0	- 0.1

Table 2: Competing linear classifiers for `compas_arrest`, and their predictions on a person with features x_p . We show the baseline model that we would deploy (left) and the models trained to evaluate individual ambiguity (middle) and maximum discrepancy (right). Here, the baseline model predicts $h(x_p) = +1$ while the other models predict $h(x_p) = -1$. As shown, there exists at least two models with only 1% additional training error that predict that x_p would not recidivate. Moreover, the coefficients and importance among the models changes. Models in the 1%-level set exhibit an overall ambiguity of 44%, meaning that 44% of individuals (2355 individuals) could have been assigned a different prediction by a model with comparable error.

- Dealing with multiplicity is complicated, and common tie-breaking strategies (e.g. by secondary objectives such as fairness) sometimes fail to resolve multiplicity.

On the Incidence of Multiplicity. We find that predictive multiplicity arises often in recidivism prediction tasks. In the `compas_arrest` dataset with an error tolerance of 1%, we find that 2,355 individuals (44%) were given a different prediction by some alternative model. For the `compas_violent` dataset with the same error tolerance, 53% of individuals could have been given a different prediction. This means that *the majority of recidivism predictions made by the optimal model are dependent on how the model is chosen for deployment*. Across the 9 prediction tasks we study, we find that between 4% and 53% of individuals are given ambiguous predictions, depending on the dataset.

Furthermore, for both `compas_arrest` and `compas_violent` we find a *single alternative model* in the 1% level set that gives 17% of individuals a different prediction. This multiplicity generalizes to discrepancy in the test set as well, as shown in Figure 3. See Table 2 for an example of the alternative models we train and their effects on an individual prediction. See Figure 3 for a summary of predictive multiplicity in each prediction task we study. See Appendix B for a profile of predictive multiplicity for all datasets.

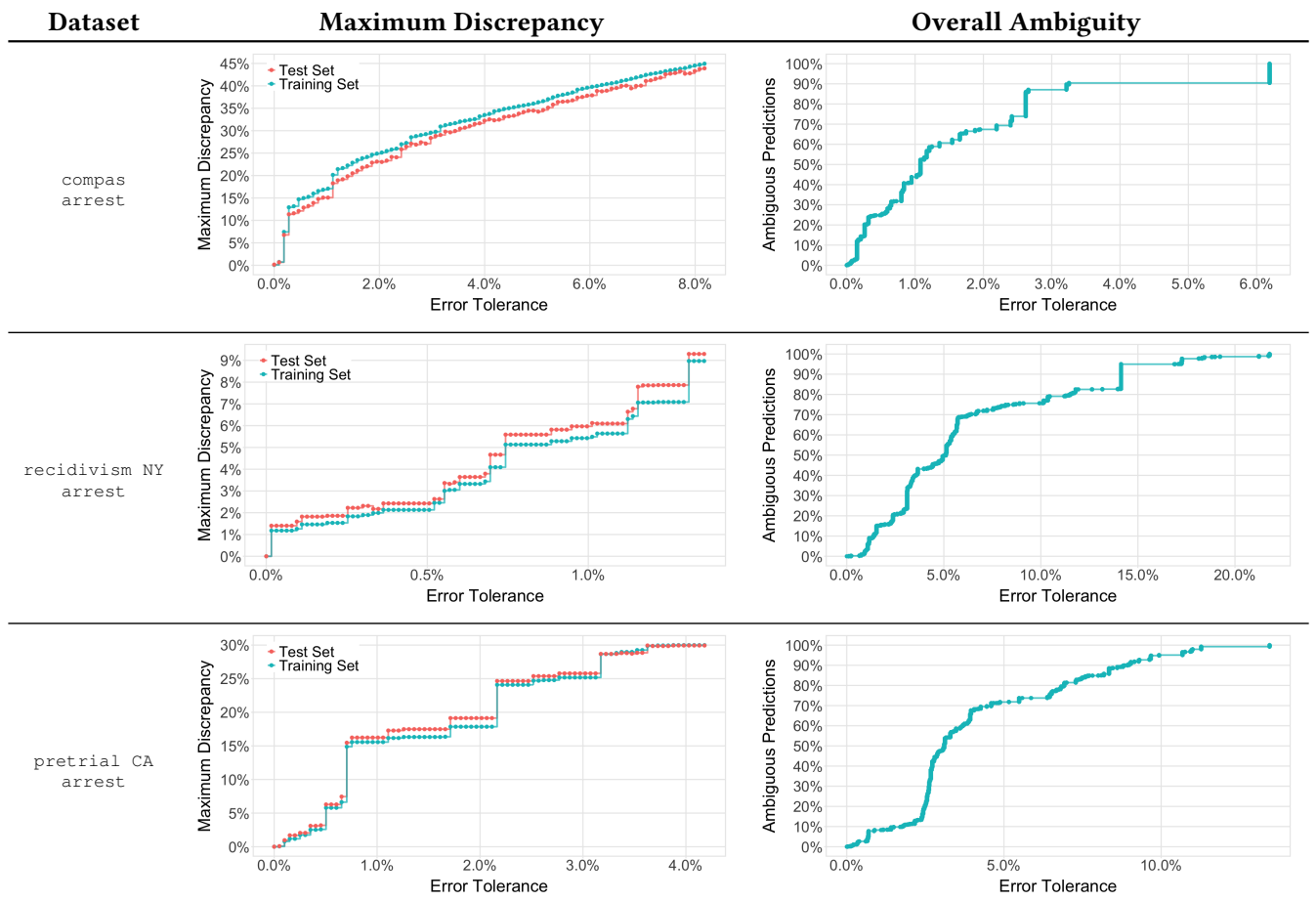
For each prediction task we consider, the baseline model performs well and does not overfit (see Table 1). Still, we observe that

the baseline model, which optimizes training error, does not necessarily optimize test error. For example, in `compas_arrest`, the model that optimizes test error is only in the set of good models when $\epsilon \geq 1\%$. This shows why we should consider the set of good models, and not only the model(s) which exactly minimize training error. Our results show why in practice, it is not defensible to rule out competing models that have wildly different predictions on the basis of small differences in training error. We see that even when we produce baseline models that generalize well, the baseline model is not necessarily the model that optimizes our actual objective, error in deployment. We report results for an error tolerance of $\epsilon = 1\%$ as a conservative default.

On Measurement with Standard Tools. Our results show that multiplicity can arise even in settings where we train classifiers using an objective function that is guaranteed to produce a unique optimal solution.⁸ These findings illustrate how evaluating and reporting multiplicity can reveal useful information even without specialized tools. In `compas_arrest`, for example, an ad hoc analysis of multiplicity among models that achieve an error within 1% of optimal reveals an estimate of ambiguity of 10% and a maximum discrepancy of 7% (i.e., 10% of predictions can be flipped by any “good model”, and 7% can be flipped by a single “good model”). We observe that

⁸The objective is strongly convex since all instances, even Lasso, include a tiny ℓ_2 penalty.

MEASUREMENT WITH OUR TOOLS



MEASUREMENT WITH STANDARD TOOLS

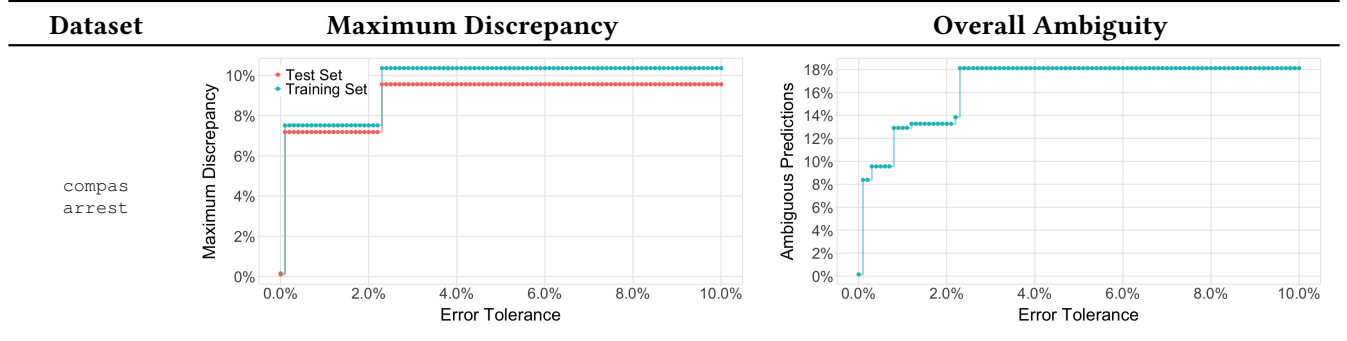


Figure 3: Overview of predictive multiplicity measures obtained with our tools (top) and using standard tools (bottom). We plot the values of maximum discrepancy (left) and overall ambiguity (right) computed over the ϵ -level set (i.e., the set of models whose training error is within ϵ of the optimal training error). Our analysis for `compas_arrest` shows that the set of models for $\epsilon = 1\%$ exhibit a maximum discrepancy of 17%, and an overall ambiguity of 44%. This means that one can change 17% of predictions by switching the baseline model with a model that is only 1% less accurate. Ambiguity reflects the fact that 44% of predictions are susceptible to model selection within this set. We observe that the values of discrepancy and ambiguity are similar on both the training and test set. We include results for other datasets in Appendix B.

these estimates are far less than those produced using our tools (44% and 17% respectively). Here, the measures of multiplicity are lesser since we have a different baseline model and an incomplete set of “good” models (i.e., we do not consider the full set of linear classifiers in the 1%-level set; only those that can be found by varying ℓ_1 and ℓ_2 penalty settings in penalized logistic regression. These results show that standard approaches for generating alternative models (e.g. varying hyperparameters) are helpful to confirm the presence of predictive multiplicity, but cannot be used to guarantee the absence of multiplicity.

On Choosing between Competing Models. When presented with a set of near equivalent models, a natural solution is to choose among them to optimize other goals. We support this practice in cases where secondary objectives are justified (see Section 5 for a discussion). Nevertheless, we note that tie-breaking does not always yield a unique model. For the `compas_arrest` dataset we find 102 unique models (i.e. distinct predictions) that differ from the optimal model by less than 1% train error and 1% disparity between African-American individuals and individuals of other ethnicities. Thus, there are cases where secondary objectives fail to sufficiently distinguish between good models.

On the Burden of Multiplicity. We find that the incidence of multiplicity can differ significantly between subgroups. In `compas_violent`, for example, the proportion of individuals whose predictions are ambiguous within the 1% level set is: 72.9% among African-American individuals ($n = 4,042$), 37.2% among Caucasian individuals ($n = 3,923$), 32.5% among Hispanic individuals ($n = 579$), and 34.4% among Other individuals ($n = 224$). In this case, we find that multiplicity disproportionately affects African-Americans compared to individuals of other ethnic groups. Groups with a larger burden of multiplicity are more vulnerable to model selection, and more likely to be affected by the ignorance of competing models.

5 CONCLUDING REMARKS

Predictive multiplicity can arise in many real-world problems, and has startling implications for the way that we choose and interact with predictive models. In applications such as recidivism prediction and credit scoring, predictive multiplicity implies irreconcilable differences in individual predictions. Thus, when a prediction problem exhibits severe predictive multiplicity, we need to carefully justify our decision to choose one model over other models that perform equally well. If there exists different models that produce conflicting predictions on individuals, and no justification to support one model over another, then it may be a better choice to forgo deployment entirely, or to assign favorable predictions to individuals who receive ambiguous predictions.

Even as there exists several techniques to choose between competing models (see Section 1.1), we do not advocate for a general prescription to “resolve” predictive multiplicity. Instead, we argue that one should measure and report multiplicity in the same way that we measure and report other kinds of model statistics [18, 31, 38, 39]. In this way, problems that exhibit predictive multiplicity can be handled on a case-by-case basis, and in a way that

allows for input from stakeholders [as per the principles of contestable design 17, 22]. The measures and machinery in this work provide a first step towards this goal.

Our proposed approach, which mirrors Breiman’s recommendations against the use of “automatic methods for feature selection,” [7], presents a series of technical and procedural benefits. Real-world problems can exhibit predictive multiplicity due to a host of reasons (e.g., the use of feature selection, training data that has missing features or latent groups, or a poorly specified model class). The process of evaluating predictive multiplicity may guide us in choosing a suitable solution. In Figure 1, for example, all models perform equally well and tie-breaking between models on the basis of out-of-sample accuracy would not achieve the goal of improving out-of-sample accuracy. Likewise, evaluating predictive multiplicity requires a systematic way to search for and evaluate alternative models, which can lead to improvements in the model in deployment, or provide safeguards against a malicious practitioner.

One potential approach to handle predictive multiplicity is to circumvent difficult choices between competing models by adapting the training process to provide a forcibly unique solution. For example, by incorporating unnecessary constraints to the problem, or by choosing an algorithm that is guaranteed to produce a single model. If such changes are only motivated by the need to produce a unique model, rather than to achieve other desirable objectives, then this would effectively relegate model selection to the training process – and thereby “automate” model selection in an unwarranted way.

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A OMITTED PROOFS

Proof of Proposition 2.1. We proceed by a simple counting argument. First, note that in the context of predictive multiplicity we consider unique models to be models which differ in at least one observed prediction. Now, there are $\binom{n}{\hat{R}(\hat{h})}$ ways to choose $\hat{R}(\hat{h})$ instances on which to make a mistake. Denote the effective lower and upper bounds of the ϵ -level set by $\underline{\epsilon} = \max(-\epsilon, -\hat{R}(\hat{h}))$ and $\bar{\epsilon} = \min(\epsilon, n - \hat{R}(\hat{h}))$ respectively, since the number of mistakes of any model must be in the interval $[0, n]$. Since the ϵ -level set is comprised only of models making a number of mistakes in the interval $[\hat{R}(\hat{h}) + \underline{\epsilon}, \hat{R}(\hat{h}) + \bar{\epsilon}]$, we see that there are at most $\sum_{i=\underline{\epsilon}}^{\bar{\epsilon}} \binom{n}{\hat{R}(\hat{h})+i}$ unique prediction vectors a model in the ϵ -level set could generate. Thus, this supplies an upper bound on the number of unique models in the ϵ -level set. Note that the number of these prediction vectors that will be achievable will vary between model classes. $\square$

Proof of Proposition 2.2. We will use the Triangle Inequality to bound the distance between the vector of predictions of the optimal model and the predictions of an alternative model in the ϵ -level set. Let $y = \{y_i\}_{i=1}^n$ be the vector of labels, let $\hat{y} = \{\hat{h}(x_i)\}_{i=1}^n$ be the vector of predictions of the optimal model, and let $y' = \{h'(x_i)\}_{i=1}^n$ be the predictions of an alternative model h' in the ϵ -level set. Note that $y, y', \hat{y} \in \{+1, -1\}^n$. Now, we can express the risk of each model, $\hat{R}(\hat{h})$ and $\hat{R}(h')$, as well as the discrepancy between the two models, $\delta(\hat{h}, h')$, in terms of these three vectors by

$$\begin{aligned}\hat{R}(\hat{h}) &= \frac{1}{4} \|y - \hat{y}\| \\ \hat{R}(h') &= \frac{1}{4} \|y - y'\| \\ \delta(\hat{h}, h') &= \frac{1}{4} \|y' - \hat{y}\|\end{aligned}$$

Next, consider the triangle formed in $\mathbb{R}^n$ by the points y, y' and $\hat{y}$. This triangle has side lengths $\|y - \hat{y}\|$, $\|y' - \hat{y}\|$ and $\|y - y'\|$. The Triangle Inequality then gives us that

$$\|y' - \hat{y}\| \leq \|y - y'\| + \|y - \hat{y}\|.$$

Substituting using the three equations above, we have

$$\delta(\hat{h}, h') \leq \hat{R}(\hat{h}) + \hat{R}(h').$$

Since $h' \in S_\epsilon(\hat{h})$, we have by the definition of the ϵ -level set that $\hat{R}(h') \leq \hat{R}(\hat{h}) + \epsilon$. We can then rewrite the above expression to yield

$$\delta(\hat{h}, h') \leq 2\hat{R}(\hat{h}) + \epsilon$$

Recall that $\delta_\epsilon := \max_{h' \in S_\epsilon(\hat{h})} \delta(\hat{h}, h')$. Since each $h' \in S_\epsilon(\hat{h})$ satisfies $\delta(\hat{h}, h') \leq 2\hat{R}(\hat{h}) + \epsilon$, we have the result that $\delta_\epsilon \leq 2\hat{R}(\hat{h}) + \epsilon$. $\square$

B SUPPORTING EXPERIMENTAL RESULTS

Dataset	n	d	Outcome Variable
compas_arrest	5,380	18	rearrest for any crime
compas_violent	8,768	18	rearrest for violent crime
pretrial_CA_arrest	9,926	22	rearrest for any crime
pretrial_CA_fta	8,738	22	failure to appear
pretrial_NY_fta	730	22	failure to appear
recidivism_CA_arrest	114,522	20	rearrest for any crime
recidivism_CA_drug	96,664	20	rearrest on drug crime
recidivism_NY_arrest	31,624	20	rearrest for any crime
recidivism_NY_drug	27,526	20	rearrest for drug crime

Table 3: Overview of recidivism prediction datasets considered in Section 4. All datasets are publically available. We include a copy of `compas_violent` and `compas_arrest` with our submission. The remaining datasets must be requested from ICPSR due to privacy restrictions.

MEASUREMENT WITH OUR TOOLS

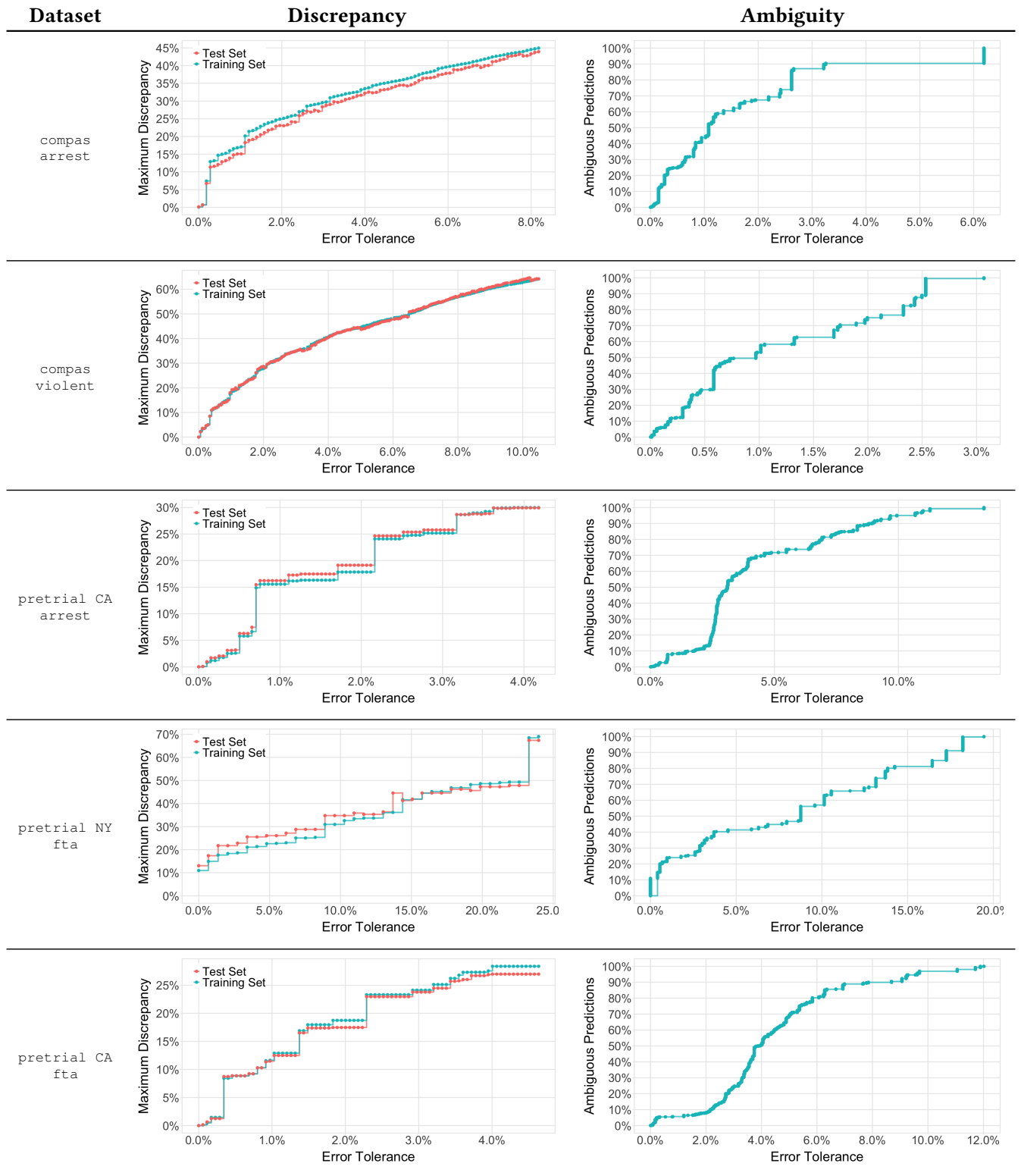


Figure 4: Multiplicity profiles for the `compas` and `pretrial` datasets. The maximum percent of conflicts between the optimal model and any model within the error tolerance (maximum discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (overall ambiguity) is shown on the right.

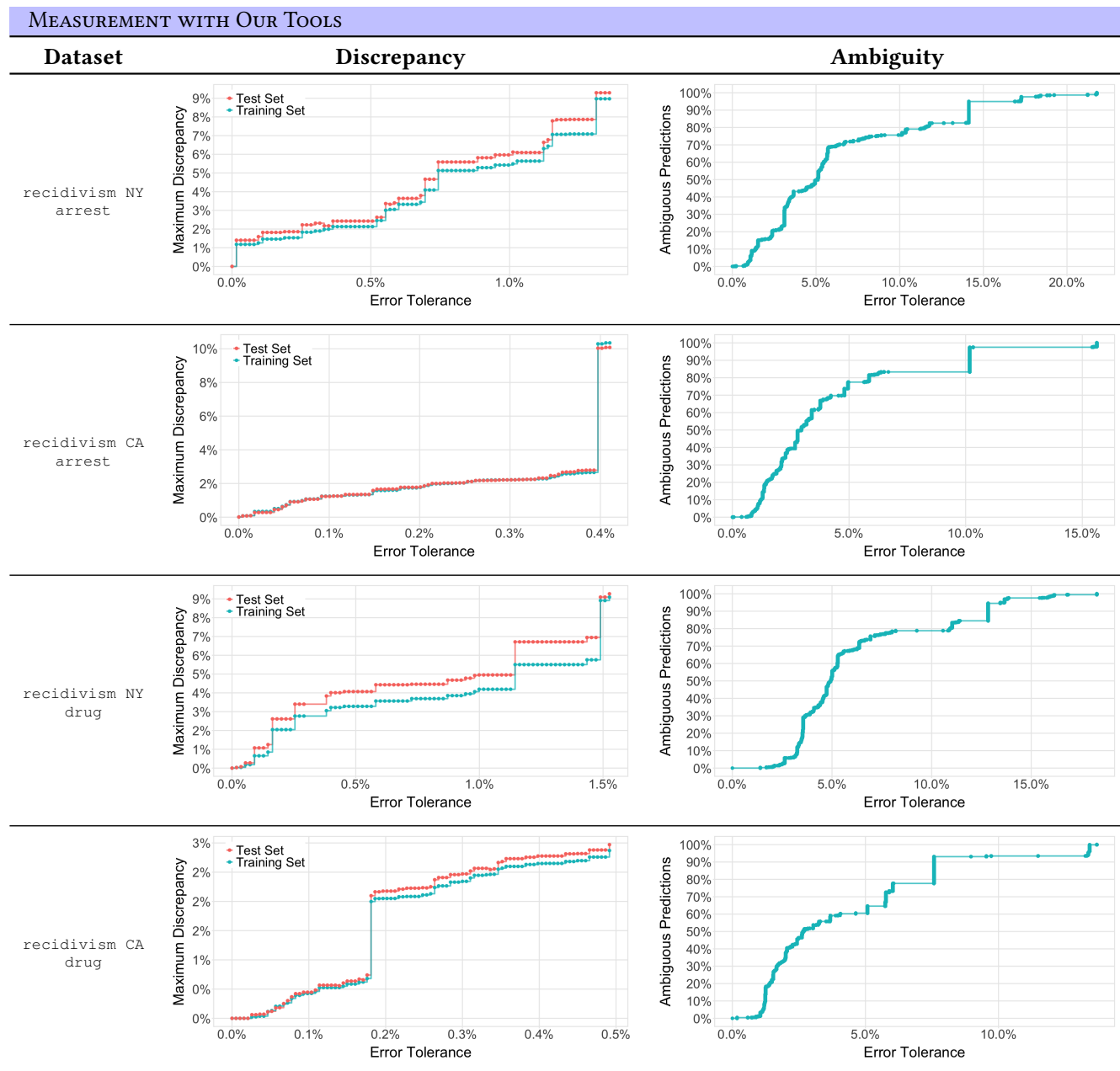


Figure 5: Multiplicity profiles for the `recidivism` datasets. The maximum percent of conflicts between the optimal model and any model within the error tolerance (maximum discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (overall ambiguity) is shown on the right.

MEASUREMENT WITH STANDARD TOOLS

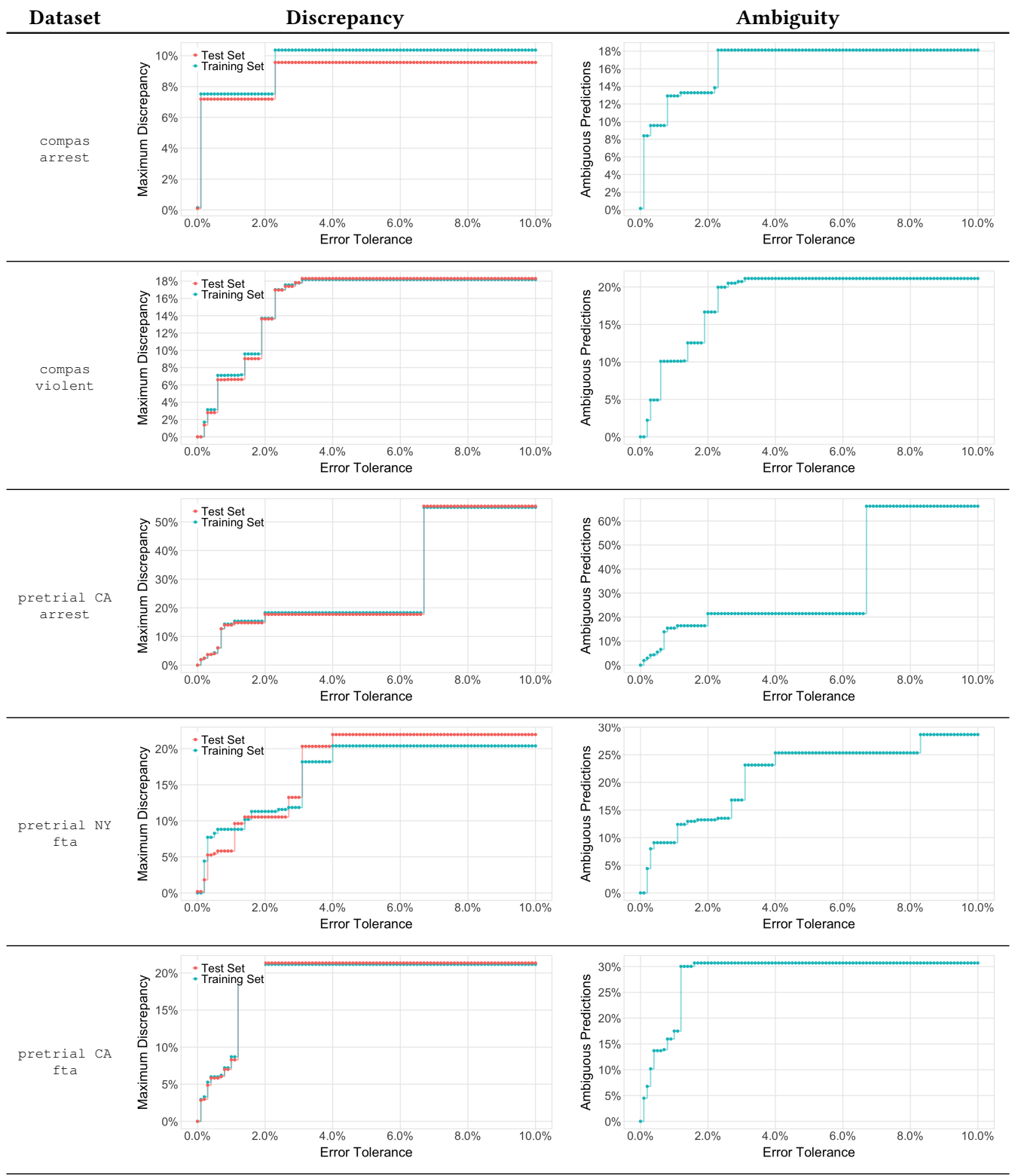


Figure 6: Multiplicity profiles for the `compas` and `pretrial` datasets, using standard methods (ℓ_1 and ℓ_2 penalty grid) to generate the level sets. The maximum percent of conflicts between the optimal model and any model within the error tolerance (maximum discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (overall ambiguity) is shown on the right.

MEASUREMENT WITH STANDARD TOOLS

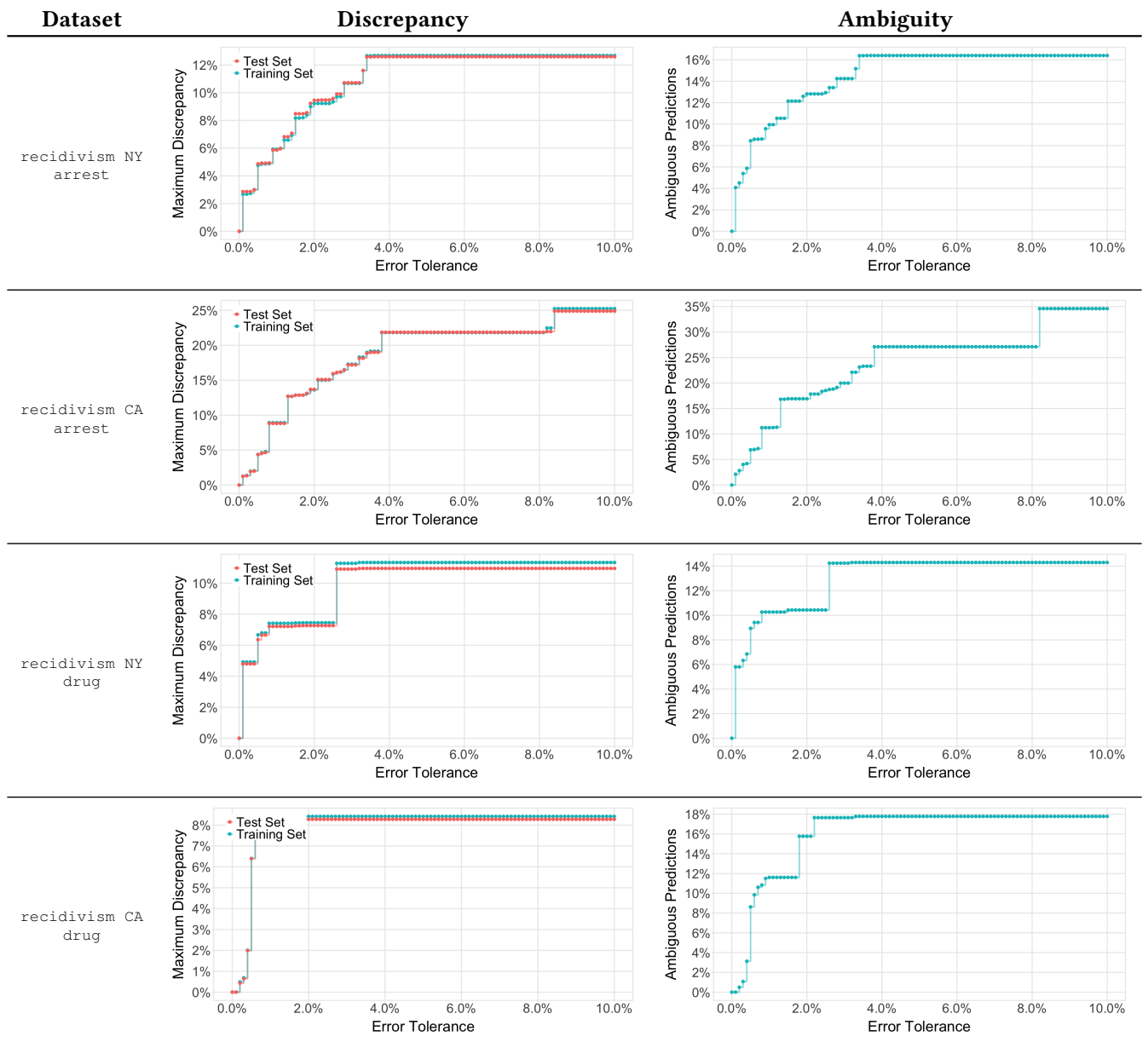


Figure 7: Multiplicity profiles for the `recidivism` datasets, using standard methods (ℓ_1 and ℓ_2 penalty grid) to generate the level sets. The maximum percent of conflicts between the optimal model and any model within the error tolerance (maximum discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (overall ambiguity) is shown on the right.